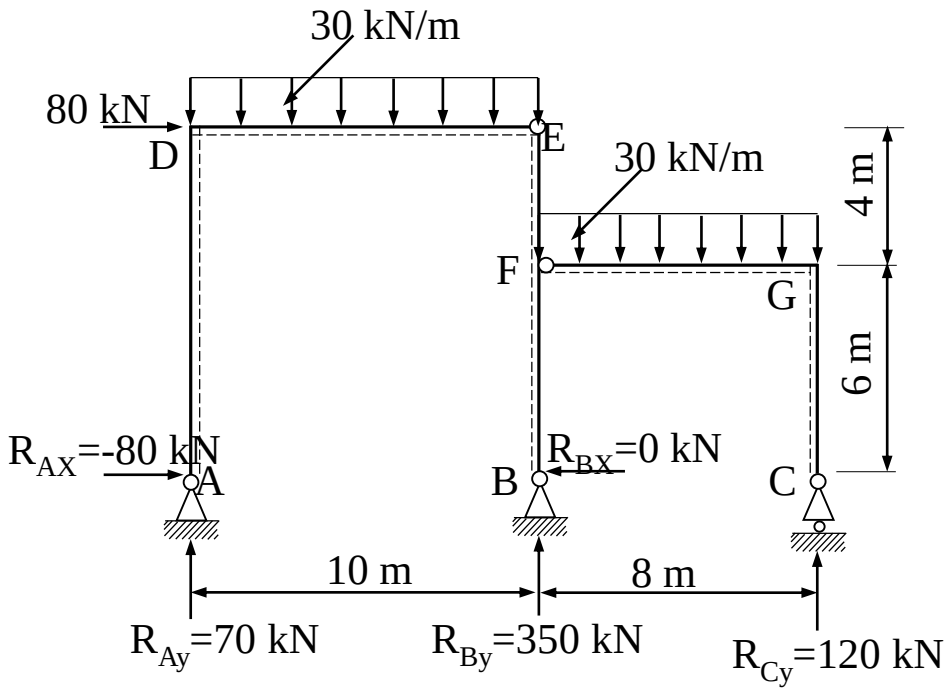
COMBINED FRAMES

Example 1:



Estimate support reactions and draw internal force diagrams of the given system.

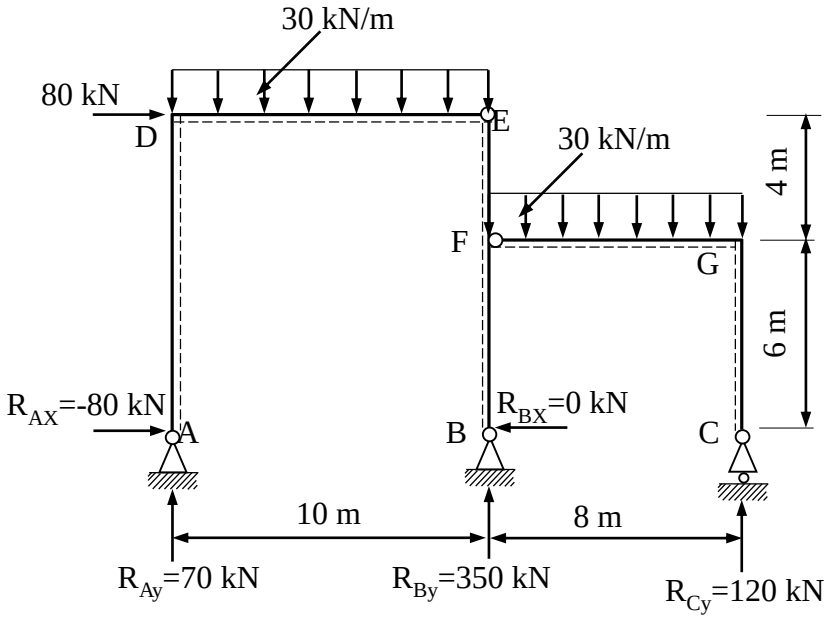
Solution:

Resultant forces of uniform distributed loads and their locations

DE member $Q_{DE} = 30 \times 10{,}0 = 300 \ \text{kN}$ $x_Q = 10{,}0/2 = 5{,}0 \ \text{m}$

FG member $Q_{FG} = 30 \times 8{,}0 = 240 \ \text{kN}$ $x_Q = 8{,}0/2 = 4{,}0 \ \text{m}$

Support Reactions



$Q_{DE} = 300$ kN

$x_Q = 5{,}0$ m

$Q_{FG} = 240$ kN

$x_Q = 4{,}0$ m

FGC system $\sum M_F = 0$ $R_{Cy} = 240 \times 4{,}0 / 8{,}0 = 120$ kN

BFEC system $\sum M_E = 0$ $R_{Bx} = (240 \times 4{,}0 - 120 \times 8{,}0) / 10{,}0 = 0$

Whole system $\sum M_B = 0$

$R_{Ay} = (-80 \times 10{,}0 + 300 \times 5{,}0 - 240 \times 4{,}0 + 120 \times 8{,}0) / 10 = 70$ kN

Whole system $\sum M_A = 0$

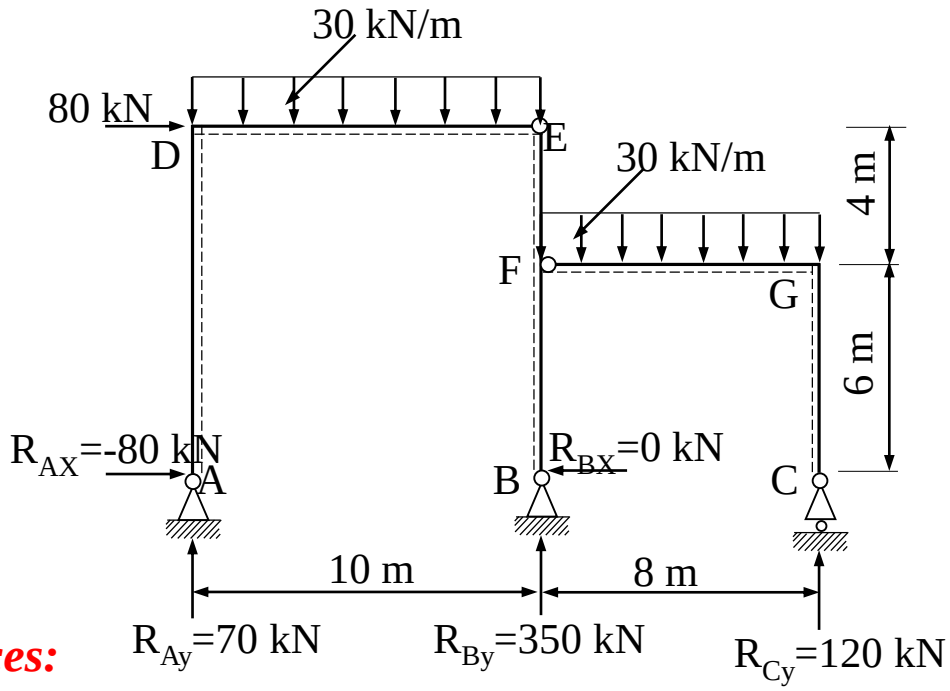
$R_{By} = (80 \times 10{,}0 + 300 \times 5{,}0 + 240 \times 14{,}0 - 120 \times 18{,}0) / 10 = 350$ kN

Check the results: $\sum P_y = 70 - 300 + 350 - 240 + 120 = 0$

ADE system $\sum M_E = 0$

$R_{Ax} = (-300 \times 5{,}0 + 70 \times 10) / 10{,}0 = -80$ kN

Check the results: $\sum P_x = -80 + 80 = 0$



Internal Forces:

AD member

$N_{AD} = N_{DA} = -70 \text{ kN} \quad V_{AD} = V_{DA} = 80 \text{ kN} \quad M_{AD} = 0$

$M_{DA} = -(-80 \times 10{,}0) = 800 \text{ kNm}$

DE member

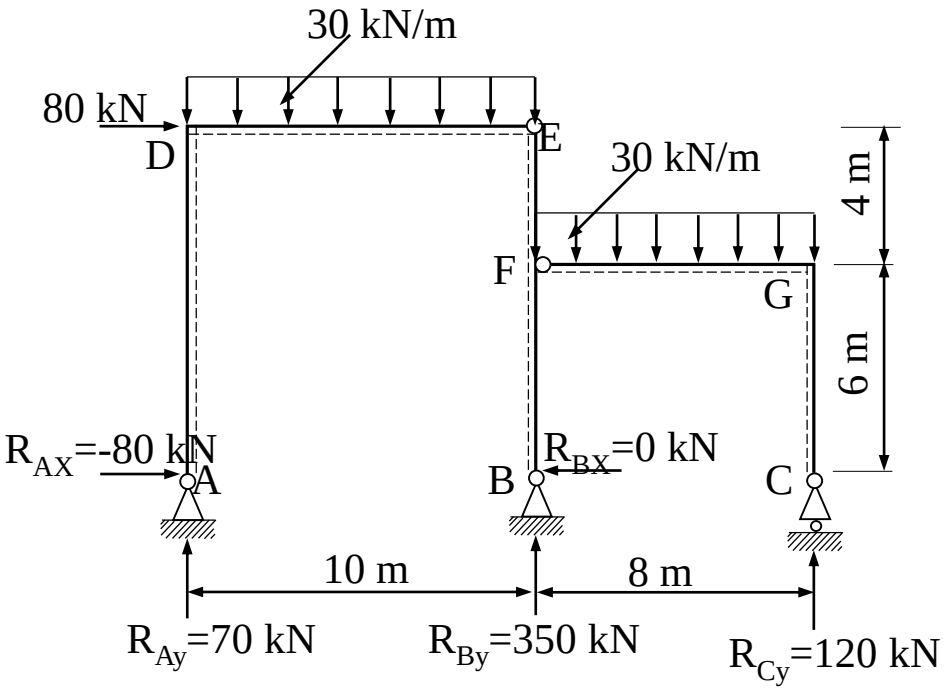
$N_{DE} = N_{ED} = 80 - 80 = 0 \quad V_{DE} = 70 \text{ kN}$

$M_{DE} = M_{DA} = 800 \text{ kNm} \quad V_{ED} = 70 - 300 = -230 \text{ kN} \quad M_{ED} = 0$

EF member

$N_{EF} = N_{FE} = 70 - 300 = -230 \text{ kN} \quad V_{EF} = V_{FE} = -80 + 80 = 0$

$M_{EF} = 0 \quad M_{FE} = 120 \times 8{,}0 - 240 \times 4{,}0 = 0$



BF member

$$N_{BF} = N_{FB} = -350 \text{ kN}$$

$$V_{BF} = V_{FB} = 0$$

$$M_{BF} = 0$$

$$M_{FB} = 0 \times 6{,}0 = 0$$

FG member

$$N_{FG} = N_{GF} = 0$$

$$V_{FG} = -120 + 240 = 120 \text{ kN}$$

$$M_{FG} = 0$$

$$V_{GF} = -120 \text{ kN} \qquad M_{GF} = 0$$

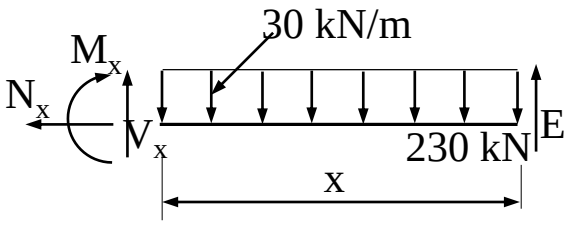
GC member

$$N_{GC} = N_{CG} = -120 \text{ kN} \qquad V_{GC} = V_{CG} = 0$$

$$M_{GC} = M_{GF} = 0 \qquad M_{CG} = 0$$

Since the sign of the shear forces at D and E ends of DE member are different, shear force is zero at any location and moment is maximum at that point.

Maximum moments:

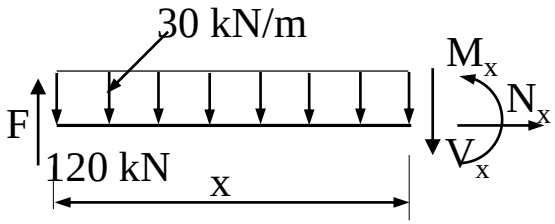


$$V_x = -230 + 30x = 0 \qquad x_0 = 230/30 = 7{,}67\,\text{m}$$

Distance of zero shear force to point D 10,0-7,67=2,33 m

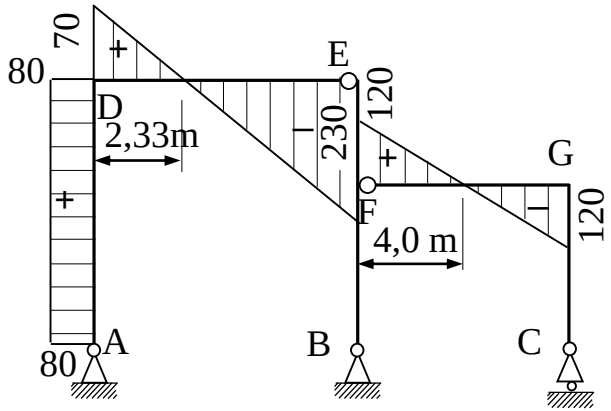
$$M_x = 230x - 30x^2/2 = 230x - 15x^2$$

$$M_{enb.} = 230 \times 7{,}67 - 15 \times 7{,}67^2 = 881{,}67 \ \text{kNm}$$

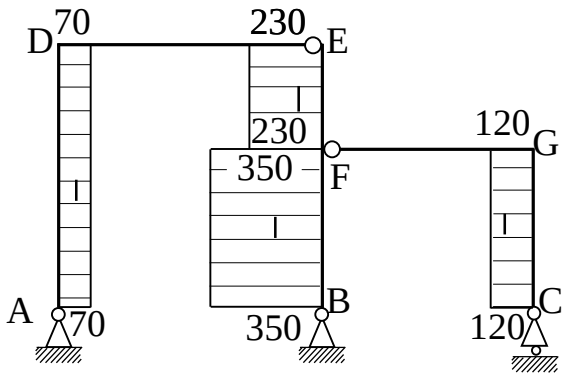


$$x_0 = V_{FG}/q = 120/30 = 4{,}0 \ \text{m}$$

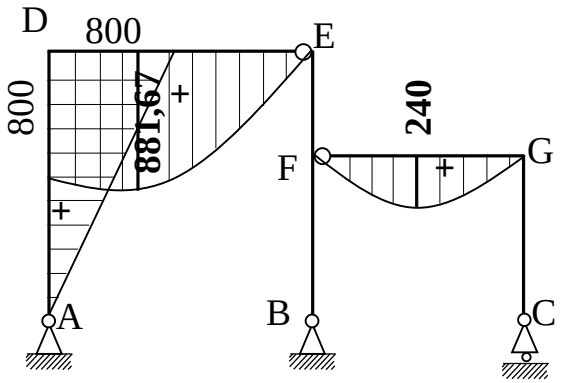
$$M_{enb.} = M_{FG} + V_{FG}^2/(2q) \longrightarrow M_{enb.} = 120^2/(2 \times 30) = 240 \ \text{kNm}$$


80
70
D
2,33m
E
230
120
G
F
120
4,0 m
A
B
C
80

V (kN)

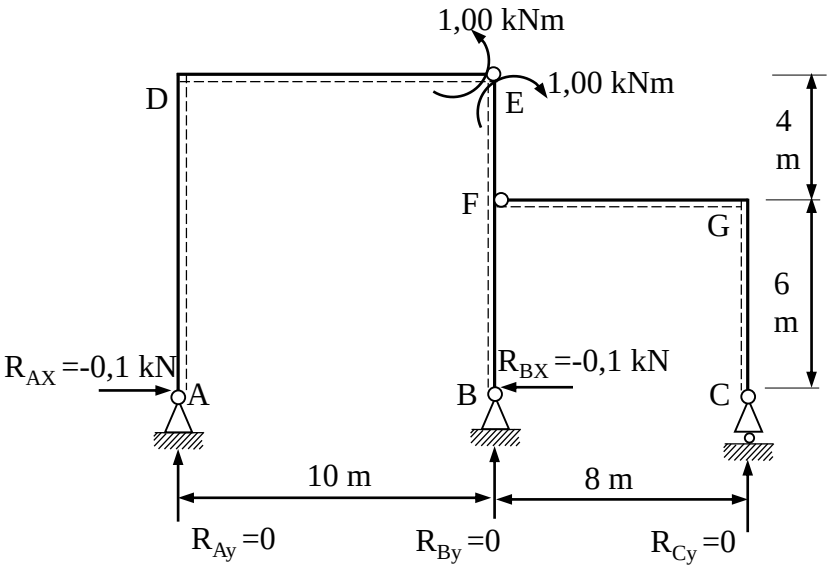

D
70
230
E
230
120
G
350
F
A
70
B
350
C
120

N (kN)


D
800
E
800
881,67
240
F
G
A
B
C

M (kNm)

Example 2:



Estimate support reactions and draw internal force diagrams of the given system.

Solution:

Support reactions

FGC system $\sum M_F = 0$ $\quad R_{Cy} = 0$

BFEC system $\sum M_E = 0$ $\quad R_{Bx} = -1/10 = -0{,}10 \ \text{kN}$

Whole system $\sum M_B = 0$ $\quad R_{Ay} = (1{,}00 - 1{,}00)/10{,}0 = 0$

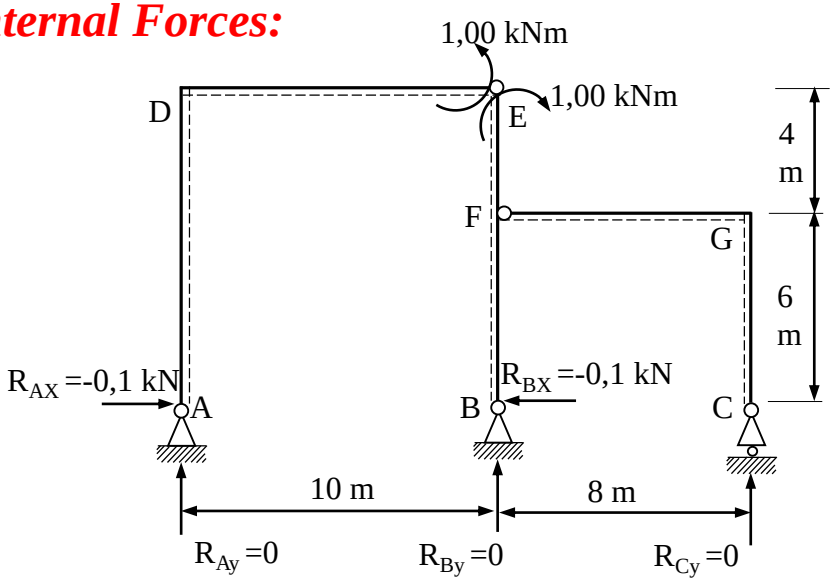
Whole system $\sum M_A = 0$ $\quad R_{By} = (1{,}00 - 1{,}00)/10{,}0 = 0$

Check the results: $\sum P_y = 0 + 0 + 0 = 0$

ADE system $\sum M_E = 0$ $\quad R_{Ax} = -1{,}00/10{,}0 = -0{,}10 \ \text{kN}$

Check the results: $\sum P_x = -0{,}10 + 0{,}10 = 0$

Internal Forces:



AD member

$$N_{AD} = N_{DA} = 0$$

$$V_{AD} = V_{DA} = 0{,}10 \ \text{kN}$$

$$M_{AD} = 0$$

$$M_{DA} = -(-0{,}10 \times 10{,}0)$$

$$M_{DA} = 1{,}00 \ \text{kNm}$$

DE member

$$N_{DE} = N_{ED} = 0{,}10 \ \text{kN} \qquad V_{DE} = V_{ED} = 0$$

$$M_{ED} = M_{DA} = 1{,}00 \ \text{kNm} \qquad M_{ED} = 1{,}00 \ \text{kNm}$$

EF member

$$N_{EF} = N_{FE} = 0 \qquad V_{EF} = V_{FE} = -0{,}10 \ \text{kN} \qquad M_{EF} = 1{,}00 \ \text{kNm}$$

$$M_{FE} = 0{,}10 \times 6{,}0 = 0{,}60 \ \text{kNm}$$

BF member

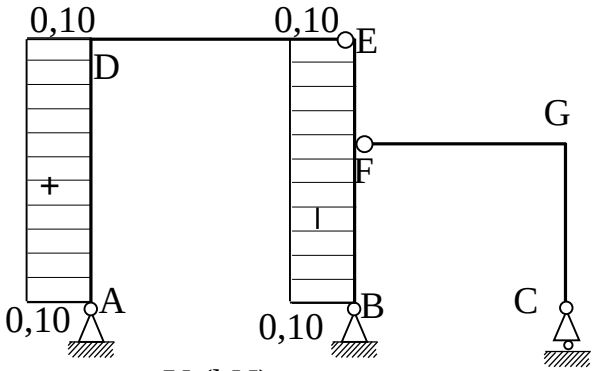
$$N_{BF} = N_{FB} = 0 \qquad V_{BF} = V_{FB} = -0{,}10 \ \text{kN} \qquad M_{BF} = 0$$

$$M_{FB} = 0{,}10 \times 6{,}0 = 0{,}60 \ \text{kNm}$$

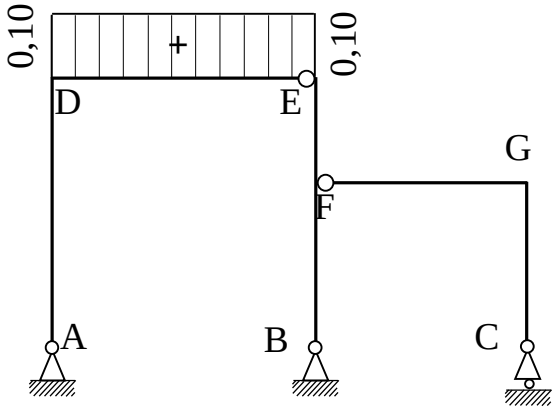
FG member

$$N_{FG} = N_{GF} = 0 \qquad V_{FG} = V_{GF} = 0 \qquad M_{FG} = 0 \qquad M_{GF} = 0$$

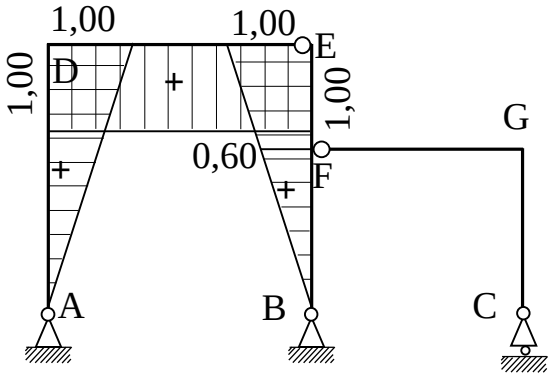
GC member $N_{GC} = N_{CG} = 0$


0,10
0,10
D
E
G
F
+
A
B
C
0,10
0,10

V (kN)

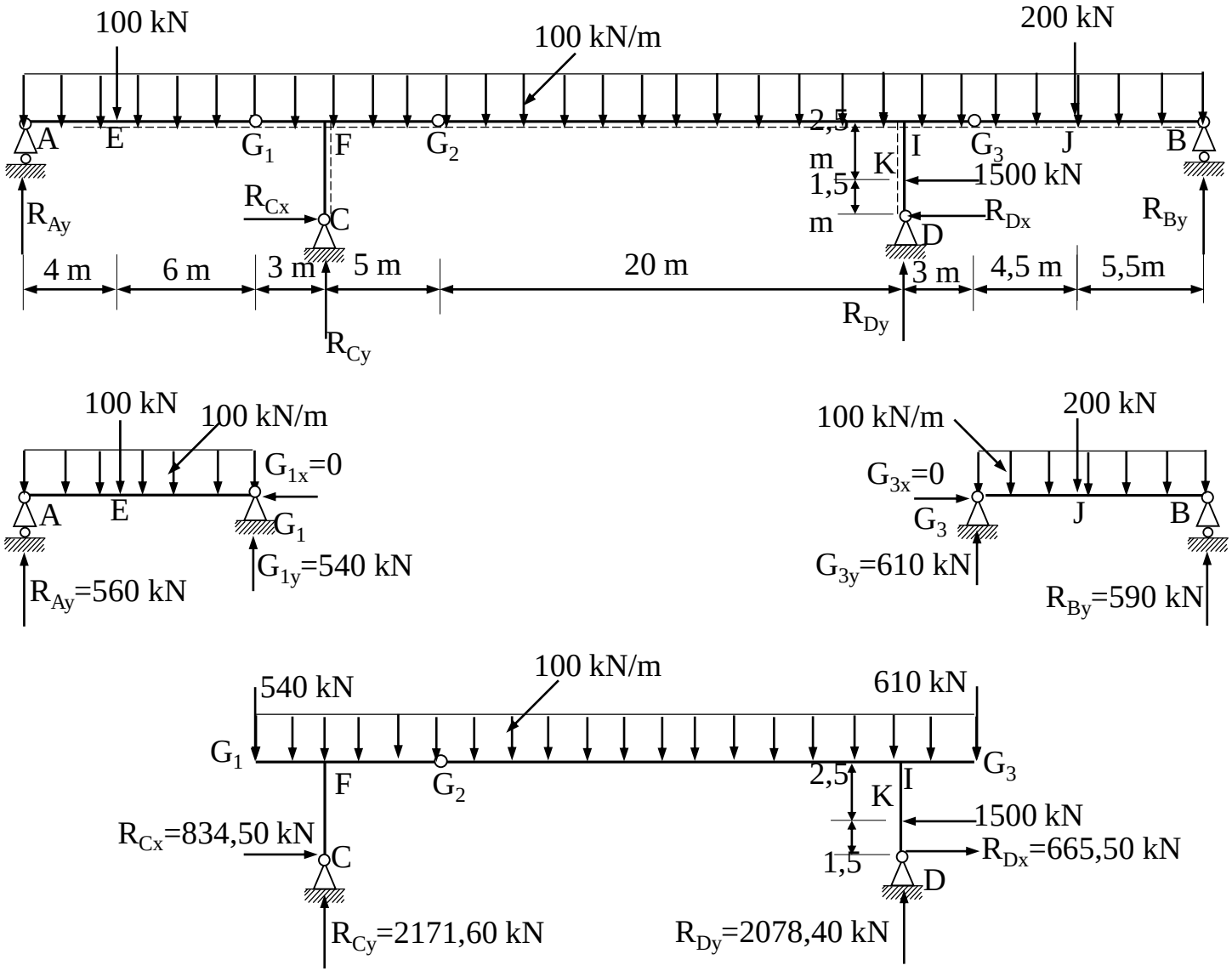

0,10
+
0,10
D
E
G
F
A
B
C

N (kN)

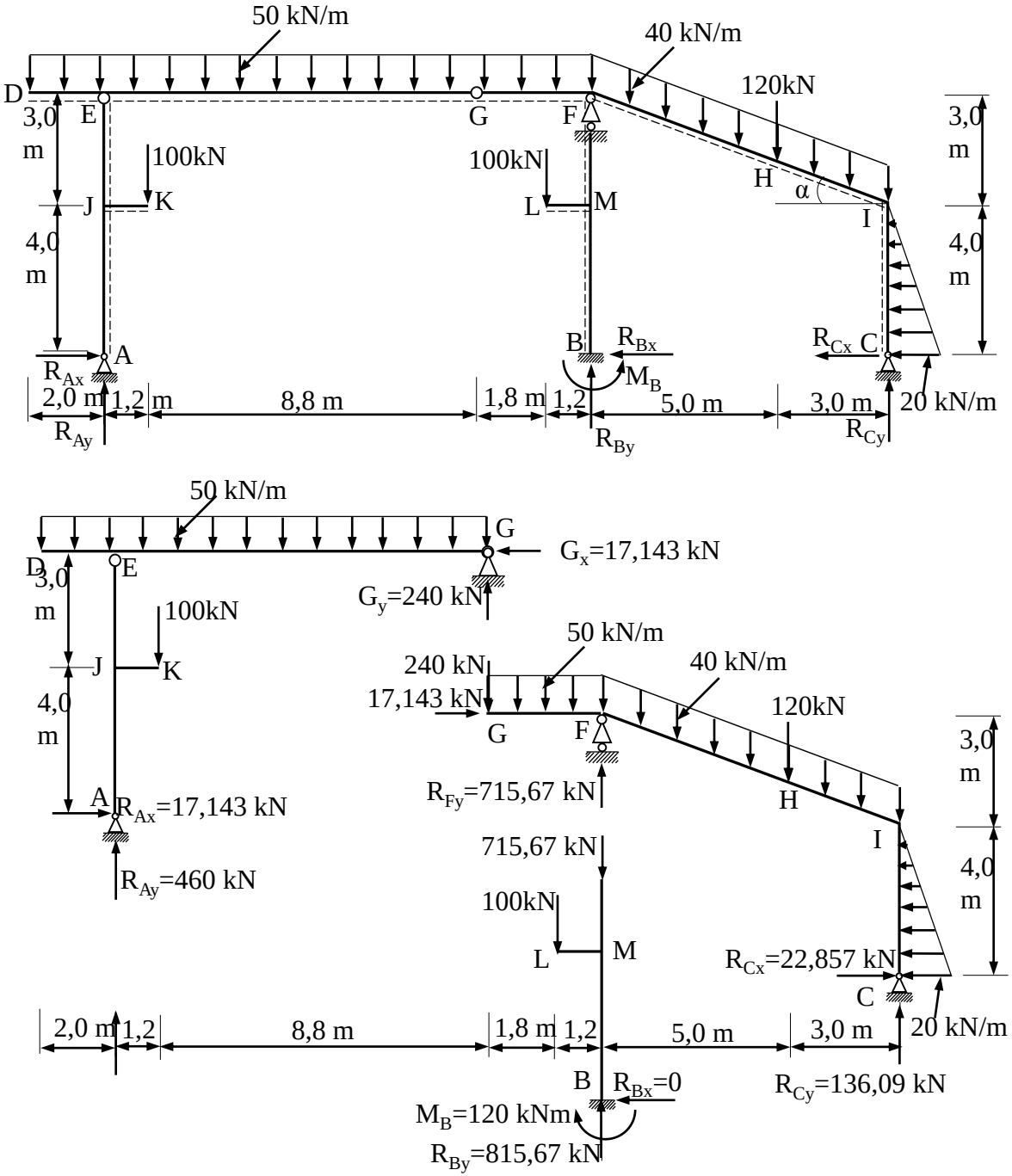

1,00
1,00
D
E
1,00
+
G
0,60
F
+
+
A
B
C

M (kNm)

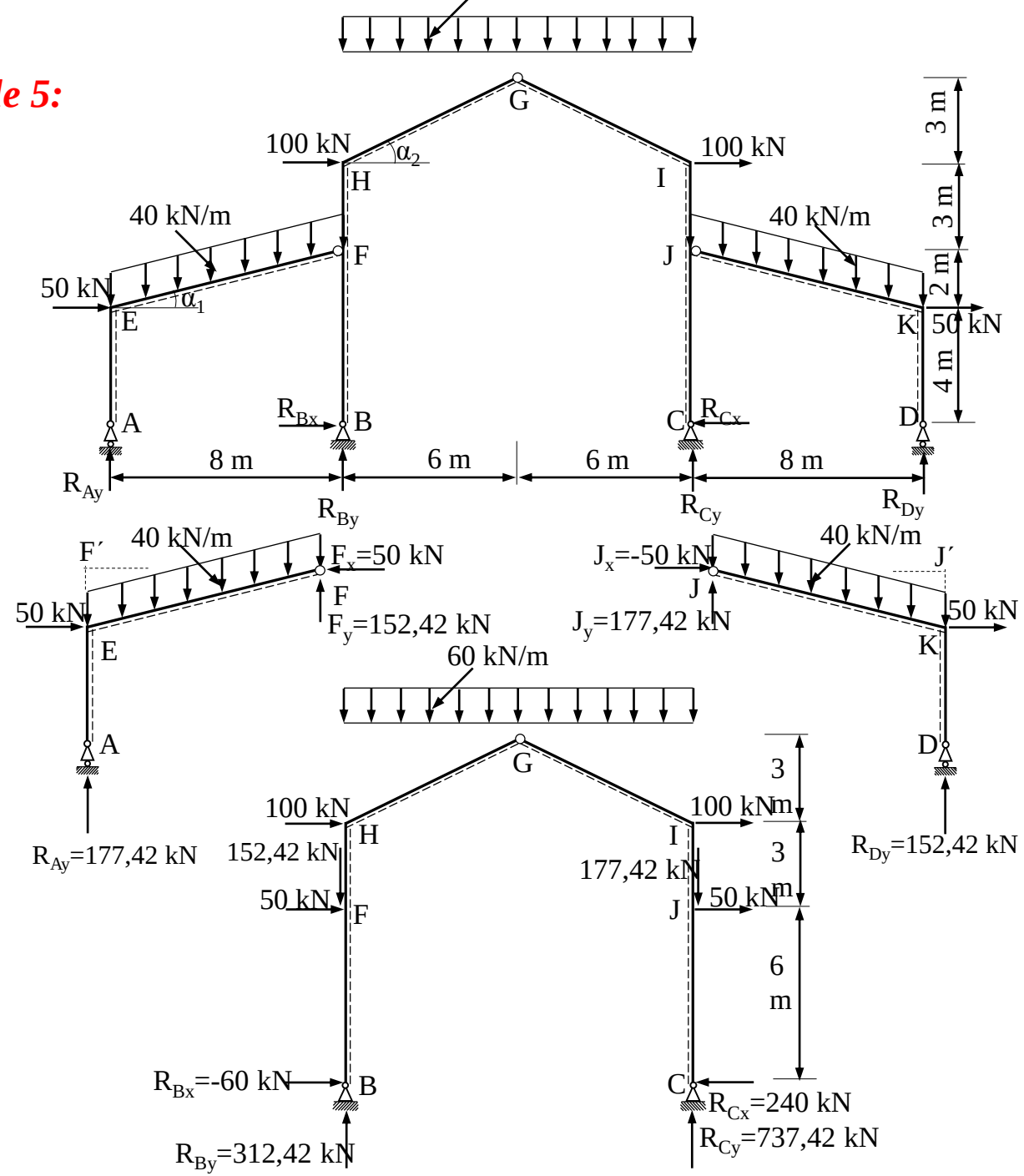
Example 3:


100 kN
100 kN/m
200 kN
A
E
G1
F
G2
I
G3
J
B
K
2,5 m
1,5 m
1500 kN
RAy
RCx
C
RDx
RBy
D
4 m
6 m
3 m
5 m
20 m
3 m
4,5 m
5,5m
RCy
RDy
100 kN
100 kN/m
G1x=0
A
E
G1
G1y=540 kN
RAy=560 kN
100 kN/m
200 kN
G3x=0
G3
J
B
G3y=610 kN
RBy=590 kN
100 kN/m
540 kN
610 kN
G1
F
G2
G3
2,5 m
K
I
1500 kN
RCx=834,50 kN
C
RDx=665,50 kN
1,5 m
D
RCy=2171,60 kN
RDy=2078,40 kN

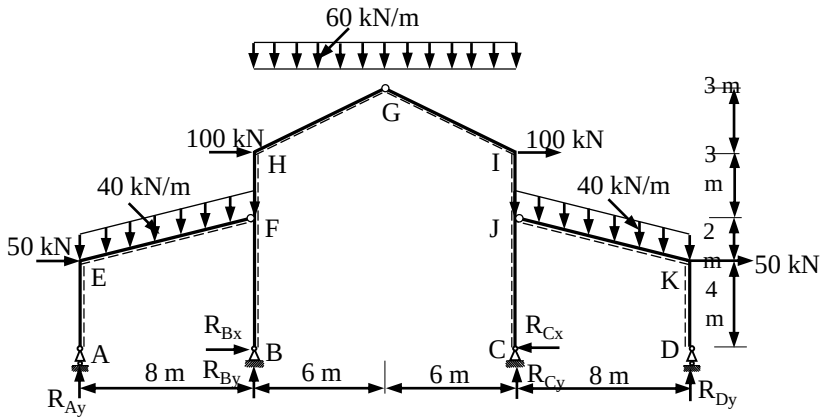
Example 4:

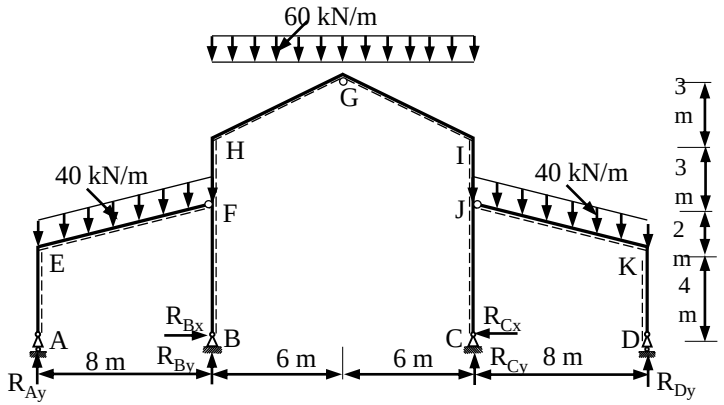

50 kN/m
40 kN/m
120kN
D
E
G
F
3,0 m
100kN
100kN
H
α
3,0 m
J
K
L
M
I
4,0 m
4,0 m
B
R_{Bx}
R_{Cx}
C
A
R_{Ax}
M_B
2,0 m
1,2 m
8,8 m
1,8 m
1,2
5,0 m
3,0 m
20 kN/m
R_{Ay}
R_{By}
R_{Cy}
50 kN/m
G
G_x=17,143 kN
D
E
3,0 m
G_y=240 kN
100kN
J
K
240 kN
50 kN/m
40 kN/m
17,143 kN
120kN
4,0 m
G
F
3,0 m
A
R_{Ax}=17,143 kN
R_{Fy}=715,67 kN
H
I
715,67 kN
4,0 m
R_{Ay}=460 kN
100kN
L
M
R_{Cx}=22,857 kN
C
2,0 m
1,2
8,8 m
1,8 m
1,2
5,0 m
3,0 m
20 kN/m
B
R_{Bx}=0
R_{Cy}=136,09 kN
M_B=120 kNm
R_{By}=815,67 kN

Example 5:

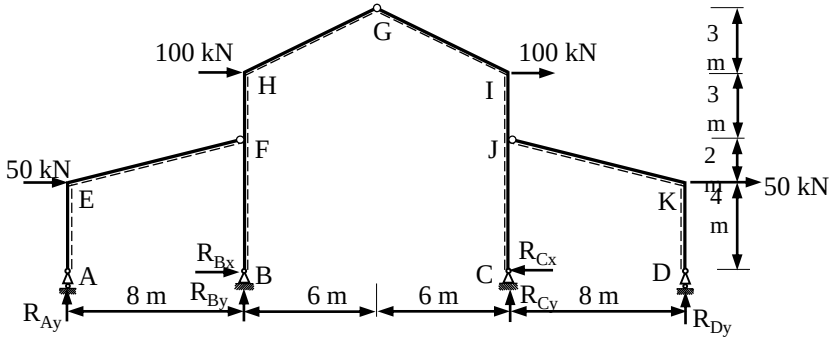

60 kN/m
G
100 kN
α2
H
I
100 kN
3 m
3 m
2 m
4 m
40 kN/m
F
J
40 kN/m
50 kN
α1
E
K
50 kN
A
RBx
B
C
RCx
D
8 m
6 m
6 m
8 m
RAy
RBy
RCy
RDy
F´
40 kN/m
Fx=50 kN
F
Fy=152,42 kN
Jx=-50 kN
J
Jy=177,42 kN
40 kN/m
J´
50 kN
E
K
50 kN
60 kN/m
A
D
G
3 m
3 m
6 m
100 kN
H
I
100 kN
152,42 kN
177,42 kN
50 kN
F
J
50 kN
RAy=177,42 kN
RDy=152,42 kN
RBx=-60 kN
B
C
RCx=240 kN
RBy=312,42 kN
RCy=737,42 kN

Example 6:



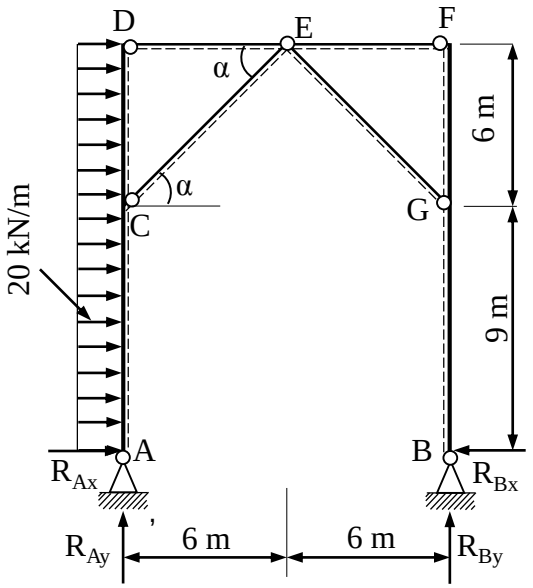


Symmetrical System



Antimetric System

Example 7:



Estimate support reactions and draw internal force diagrams of the given system.

Solution:

$$L_{CE} = \sqrt{36+36} = 8{,}485 \text{ m}$$

$$\alpha = 45^0$$

$$\cos 45 = 0{,}707 \qquad \sin 45 = 0{,}707$$

Support Reactions:

Since A and B are at same level

Whole system $\sum M_B = 0 \longrightarrow R_{Ay} = -20 \times 15{,}0 \times 7{,}5/12 = -187{,}50 \text{ kN}$

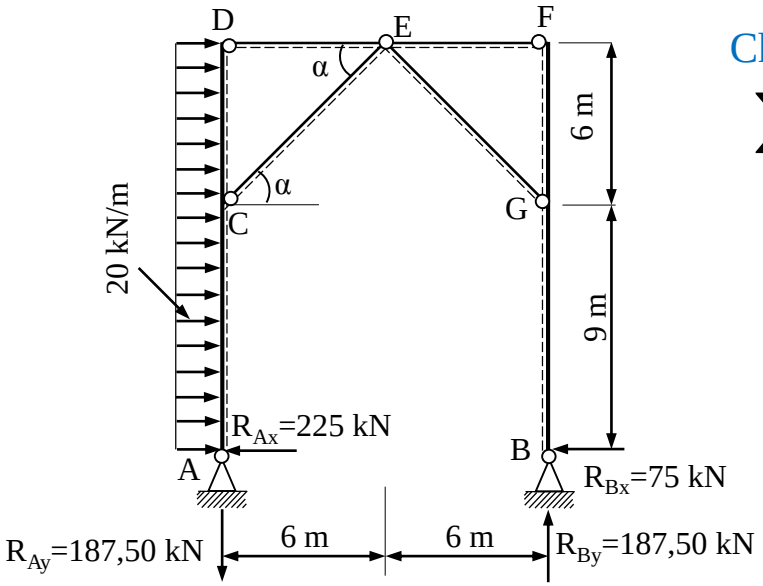
Whole system $\sum M_A = 0 \longrightarrow R_{By} = 20 \times 15{,}0 \times 7{,}5/12 = 187{,}50 \text{ kN}$

Check the results: $\sum P_y = -187{,}50 + 187{,}50 = 0$

ACDE system $\sum M_E = 0$

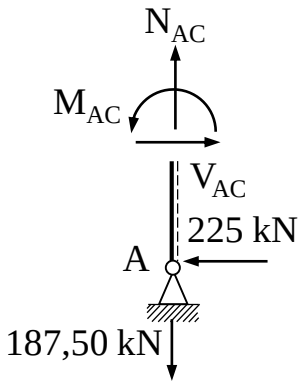
$$R_{Ax} = (-187{,}50 \times 6{,}0 - 20 \times 15{,}0 \times 7{,}5)/15 = -225{,}00 \text{ kN}$$

BGFE system $\sum M_E = 0 \longrightarrow R_{Bx} = 187{,}50 \times 6{,}0/15 = 75{,}00 \text{ kN}$



Check the results:

$$\sum P_x = -225 + 300 - 75 = 0$$



Internal Forces

Since CE, DE, EG ve EF members have hinge at two ends and there is no external loads on the members, moment and shear force do not occur in the member, only axial force exists.

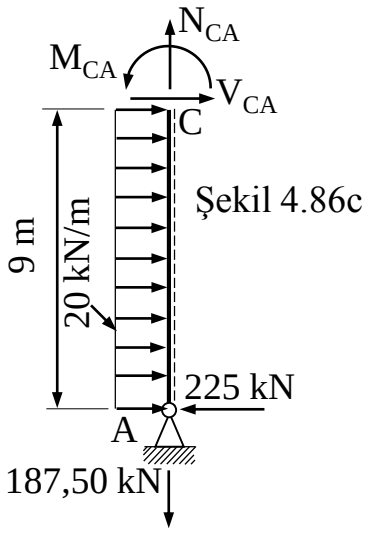
Cut from A of AC member:

$$\sum P_y = 0 \longrightarrow N_{AC} = 187{,}50 \text{ kN}$$

$$\sum P_x = 0 \longrightarrow V_{AC} = 225 \text{ kN}$$

Since support at A is a pin support:

$$M_{AC} = 0$$



Şekil 4.86c

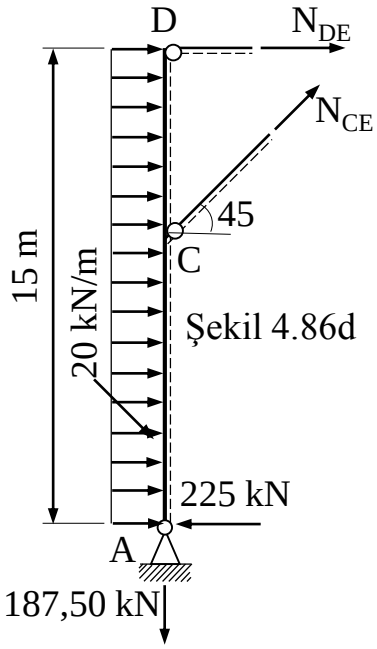
Cut from bottom of the C of AC member

$$\sum P_y = 0 \longrightarrow N_{CA} = 187{,}50 \text{ kN}$$

$$\sum P_x = 0 \longrightarrow V_{CA} = 225 - 20 \times 9{,}0 = 45 \text{ kN}$$

$$\sum M_C = 0$$

$$M_{CA} = 225 \times 9{,}0 - 180 \times 4{,}5 = 1215 \text{ kNm}$$



Şekil 4.86d

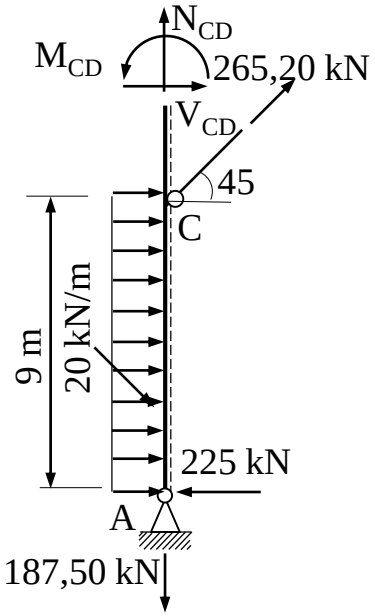
Cut from DE and CE members and consider the left part:

$$\sum P_y = 0 \longrightarrow N_{CE} \cdot \sin 45 - 187{,}5 = 0$$

$$N_{CE} = 187{,}50 / 0{,}707 = 265{,}20 \text{ kN}$$

$$\sum P_x = 0 \longrightarrow N_{DE} = 225 - 265{,}20 \times \cos 45 - 300$$

$$N_{DE} = -262{,}50 \text{ kN}$$



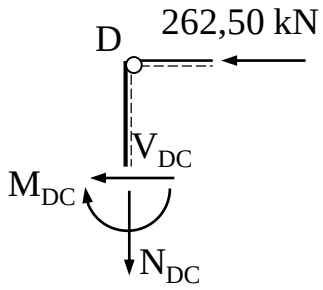
Cut from top of C of ACD system:

$$\sum P_y = 0 \longrightarrow N_{CD} = -265{,}20 \times \sin 45 + 187{,}50 = 0$$

$$\sum P_x = 0 \longrightarrow V_{CD} = 225 - 265{,}20 \times \cos 45 - 180$$

$$\longrightarrow V_{CD} = -142{,}50 \ \text{kN}$$

$$M_{CD} = M_{CA} = 1215 \ \text{kNm}$$

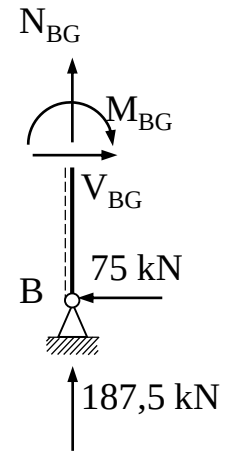


$$\sum P_y = 0 \longrightarrow N_{DC} = 0$$

$$\sum P_x = 0 \longrightarrow V_{DC} = -262{,}5 \ \text{kN}$$

$$\sum M_D = 0 \qquad M_{DC} = 0$$

Since the sign of the shear forces at two ends of AC and CD members are same, there is no zero shear force.



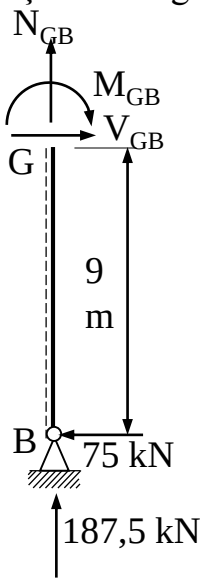
Şekil 4.86g

Cut from B of BG member

$$\sum P_y = 0 \longrightarrow N_{BG} = -187{,}50 \text{ kN}$$
$$\sum P_x = 0 \longrightarrow V_{BG} = 75 \text{ kN}$$

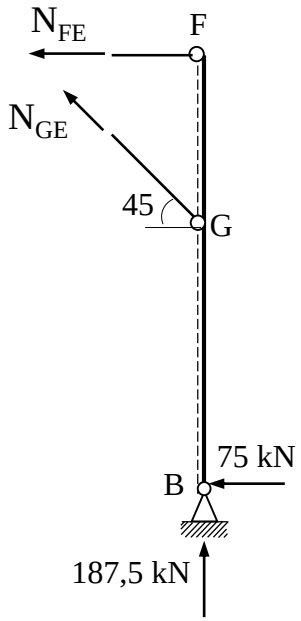
Since the support at B is a pin support

$$M_{BG} = 0$$



Cut from bottom of G of BG member

$$\sum P_y = 0 \longrightarrow N_{GB} = -187{,}50 \text{ kN}$$
$$\sum P_x = 0 \longrightarrow V_{GB} = 75 \text{ kN}$$
$$\sum M_G = 0 \longrightarrow M_{GB} = -75 \times 9{,}0 = -675 \text{ kNm}$$



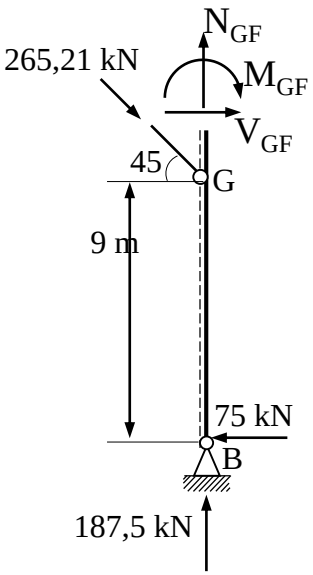
Cut from EF ve EG members of EFBG system and consider the right part:

$$\sum P_y = 0 \longrightarrow N_{GE} \cdot \sin 45 + 187{,}5 = 0$$

$$N_{GE} = -187{,}50 / 0{,}707 = -265{,}20 \text{ kN}$$

$$\sum P_x = 0 \searrow$$

$$N_{FE} = -75 - (-265{,}20 \cdot \cos 45) = 112{,}50 \text{ kN}$$



Cut from the top of the G of BF member and consider the bottom part:

$$\sum P_y = 0 \longrightarrow N_{GF} = 265{,}20 \times \sin 45 - 187{,}50 = 0$$

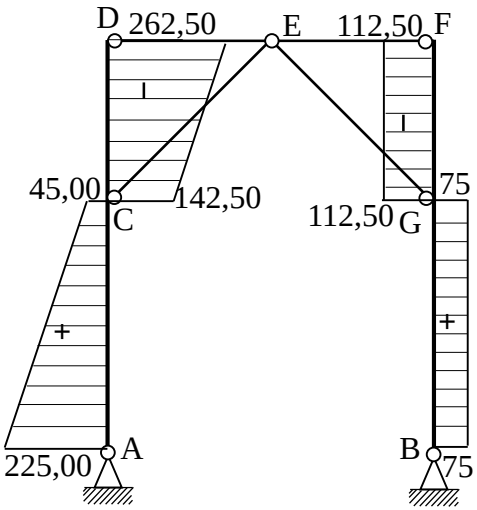
$$\sum P_x = 0 \longrightarrow V_{GF} = 75 - 265{,}20 \times \cos 45$$

$$V_{GF} = -112{,}50 \text{ kN}$$

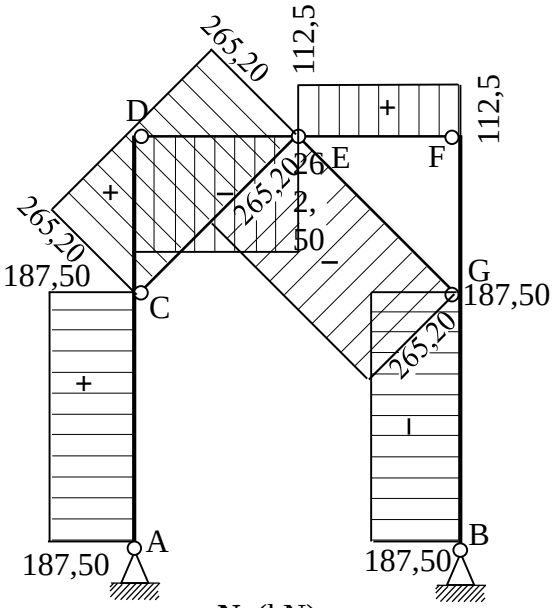
$$\sum M_G = 0 \longrightarrow M_{GF} = M_{GB} = -675 \text{ kNm}$$

$$N_{FG} = N_{GF} = 0 \qquad V_{FG} = V_{GF} = -112{,}50 \text{ kN}$$

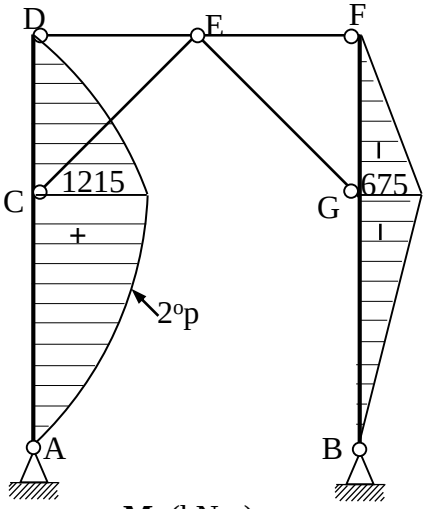
$$M_{FG} = 0$$


D
262,50
E
112,50
F
45,00
C
142,50
112,50
G
75
225,00
A
B
75

V (kN)


265,20
112,5
D
E
F
112,5
265,20
265,20
187,50
C
G
187,50
265,20
187,50
A
187,50
B

N (kN)


D
E
F
C
1215
G
675
2ºp
A
B

M (kNm)

Reference:

- Prof. İbrahim EKİZ, YAPI STATİĞİ 1, BİRSEN YAYINEVİ

Acknowledgement:

Thank you to Prof. İbrahim EKİZ for sharing this presentation with me.